

ASSIGNMENTS

2a. How many mutated genes are displayed?

There are a total of 677 genes displayed

2b. How many total mutations are present?

There are a total of 168 somatic mutations present.

2c. What gene has the highest number of mutation?

The gene with the highest number of mutation is CSMD3

2d. What is the **mutation frequency** of this gene in liver cancer (in percentage)?

For the CSMD3 gene, the frequency in this cohort is 7.14% (displayed as 1/14 cases)

2e. What is the **amino acid change** and at what **position** did it occur?

V2384Cfs*22, it started at Valine at position 2384.

2f. This is a frameshift mutation and frameshifts are almost always deleterious. They shift the reading frame of the DNA, completely changing every amino acid after the mutation and often creating a premature stop signal.

2g. This likely results in a shortened protein that cannot fold correctly, leading to a complete loss of its normal biological function.